

A2 (Conditional Independence) — Vulnerability Analysis and Fix

Trigger: Hostile review pointed out that A2 is “almost certainly false, the Achilles’ heel of the entire framework.” **Conclusion:** Diagnosis correct, but fixable. Replace with weak-correlation concentration inequality, degrade the decay rate.

1. Problem Diagnosis

Theorem 1’s proof chain:

A2 (conditional independence) $\rightarrow$ e_m independent $\rightarrow$ Hoeffding concentration $\rightarrow \exp(-2$

If expert errors have conditional correlation $\rho > 0$ (given state s): - CNN: all experts err together on blurry images - MLIP: all experts err together on transition-state configurations - Effective independent expert count: $M_{\text{eff}} \approx M/(1 + (M - 1)\rho)$

At $\rho = 0.3$, $M = 12$ degrades to $M_{\text{eff}} = 2.8$ — the exponential gain nearly vanishes.

CIFAR-10 experimental evidence: Theoretical lower bound F10.976 (assumed independence), measured F1=0.617. The gap cannot be explained by μ_s estimation error alone.

AIN experimental evidence: Global bound F10.87 matches empirical value. But this only holds when $\rho \lesssim 0.02$ — an extremely weak assumption that requires verification.

2. Fix Strategy

2.1 Weak-Correlation Hoeffding Bound

Let ρ_{ij} be the error correlation between experts i and j . With average correlation $\bar{\rho} = \frac{1}{M(M-1)} \sum_{i \neq j} \rho_{ij}$:

$$\text{Var} \left(\frac{1}{M} \sum_{i=1}^M e_i \right) = \frac{\sigma^2}{M} [1 + (M-1)\bar{\rho}] \quad (1)$$

This is an exact formula for the variance under correlation. The variance inflation factor is $1 + (M-1)\bar{\rho}$.

Crucial distinction: This variance inflation is exact for the **variance**, but plugging it into Hoeffding gives a **heuristic** bound, not a rigorous theorem. See `a2_rigorous_analysis.tex` for the rigorous treatment.

Heuristic fix: Replace M with $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$ in the Hoeffding bound:

$$F_1 \geq 1 - \frac{1}{\eta} \exp \left(-\frac{2M\Delta^2}{1 + (M-1)\bar{\rho}} \right) \quad (2)$$

Limitation of the heuristic: The MGF factorization required by Hoeffding only holds under independence. For correlated variables, this is an approximation widely used in survey sampling but not a theorem.

3. How to Estimate $\bar{\rho}$

3.1 Paired error correlation on training set

For each pair of experts (i, j) , compute their error correlation on the training set:

$$\hat{\rho}_{ij} = \text{Corr}(\mathbf{1}\{f_i(x) \neq y\}, \mathbf{1}\{f_j(x) \neq y\}) \quad (3)$$

Then average: $\hat{\bar{\rho}} = \frac{1}{M(M-1)} \sum_{i \neq j} \hat{\rho}_{ij}$.

3.2 Bootstrap confidence intervals

Use bootstrap resampling of training samples to obtain confidence intervals for $\bar{\rho}$. The effective M_{eff} can be reported with uncertainty: $M_{\text{eff}} \in [M_{\text{eff}}^{\text{low}}, M_{\text{eff}}^{\text{high}}]$ at 95% confidence.

4. When is A2 “Good Enough”?

A2 does not need to hold perfectly. It needs to hold well enough that M_{eff} remains large. The practical threshold:

- If $\bar{\rho} < 0.1$: $M = 10$ gives $M_{\text{eff}} \approx 5.3$ — still exponential gain
- If $\bar{\rho} < 0.05$: $M = 10$ gives $M_{\text{eff}} \approx 6.9$ — strong gain
- If $\bar{\rho} > 0.3$: $M = 10$ gives $M_{\text{eff}} \approx 2.7$ — gain marginal

The honest question: What is the typical $\bar{\rho}$ in real SCX deployments? For heterogeneous experts (CNN + RandomForest + XGBoost), $\bar{\rho}$ is likely low (< 0.2). For homogeneous experts (all CNNs with different initializations), $\bar{\rho}$ may be high (> 0.4). SCX must report $\bar{\rho}$ estimates alongside M_{eff} .

5. Relationship to the Rigorous Analysis

This document presents the **heuristic fix** (variance inflation plugged into Hoeffding). `a2_rigorous_analysis.tex` provides the **rigorous treatment**:

1. The Hoeffding bound with M_{eff} is **not** a theorem for correlated variables.
2. The relationship $M_{\text{eff}} = M / (1 + (M - 1)\bar{\rho})$ is exact for the variance but not for the tail probability.
3. Rigorous alternatives: Azuma-Hoeffding for martingale difference sequences, or Bernstein inequalities for weak dependence.